RAG Learning

# 3.7 Fields and Their Consequences

## 1. Gravitational Fields

**Gravitational field:** A region of space in which a mass experiences a force due to the presence of another mass. All objects with mass create a gravitational field around them. Gravity is always attractive — there is no gravitational repulsion.

### Newton's Law of Gravitation

$$F = -GMm / r^2$$

- F = gravitational force between two point masses (N)
- G = gravitational constant = $6.67 \times 10^{-11}$ N $m^2$ $kg^{-2}$
- M, m = masses of the two objects (kg)
- r = distance between the centres of mass (m)
- The negative sign indicates the force is attractive (directed inward)

This is an inverse square law: doubling the distance reduces the force by a factor of four.

### Gravitational Field Strength

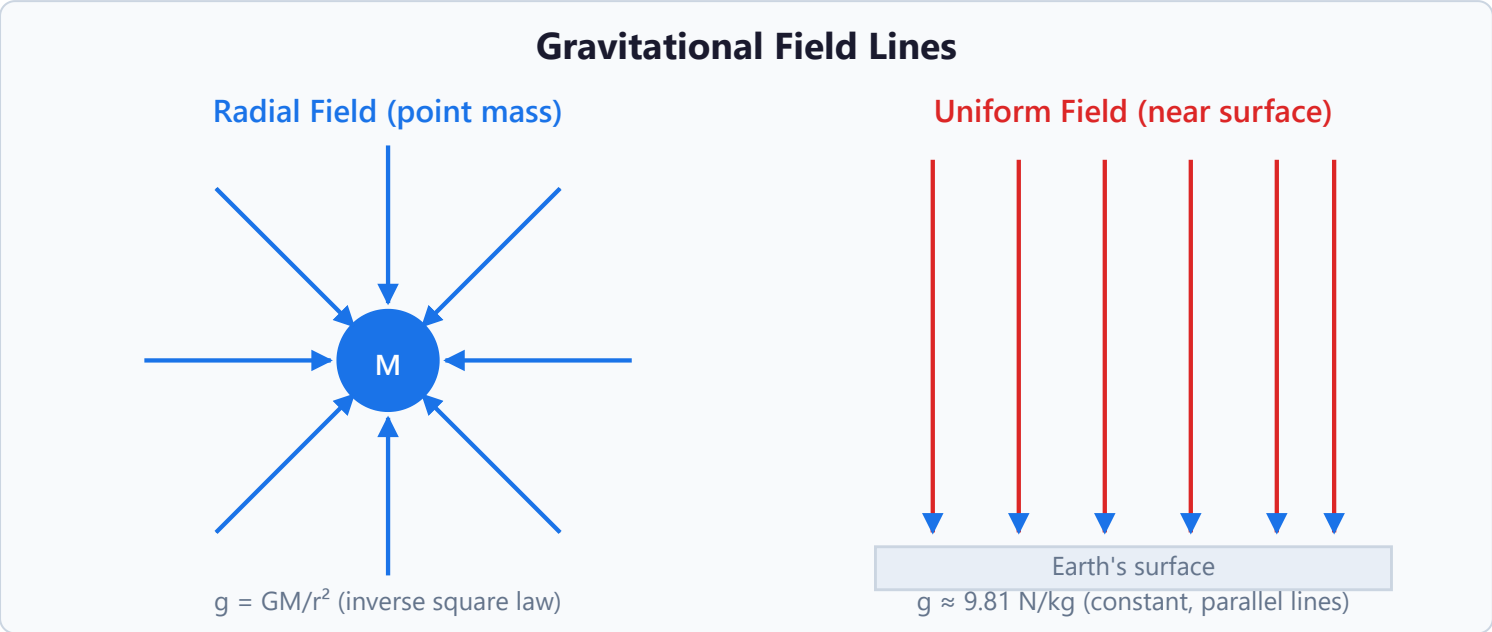
**Gravitational field strength (g):** The gravitational force per unit mass experienced by a small test mass placed at that point. g = F/m. Units: N $kg^{-1}$ (equivalently, m $s^{-2}$).

**Radial field** (point or spherical mass):

$$g = GM / r^2$$

Field strength follows an inverse square law. Field lines point radially inward towards the mass. The field is non-uniform (g decreases with distance).

**Uniform field:** Near the Earth's surface, g is approximately constant at 9.81 N $kg^{-1}$. Field lines are parallel, equally spaced, and directed downward.




## Gravitational Potential

**Gravitational potential ($V_g$):** The work done per unit mass in bringing a small test mass from infinity to that point. $V_g$ is always negative (since gravity is attractive and we define $V_g = 0$ at infinity). Units: J $kg^{-1}$.

$$V_g = -GM / r$$

**Gravitational potential energy ($E_p$):** The potential energy of a mass m at a point where the gravitational potential is $V_g$:

$$E_p = mV_g = -GMm / r$$

**The relationship between g and $V_g$:**

$$g = -dV_g/dr$$

The field strength is the negative gradient of the potential. On a $V_g$-r graph, g at any point equals the magnitude of the gradient of the tangent. Equipotential surfaces are perpendicular to field lines.

**Worked Example:** Calculate the gravitational potential and gravitational field strength at the surface of the Earth. ($M_E = 5.97 \times 10^{24}$ kg, $R_E = 6.37 \times 10^6$ m)

**Solution:**

$V_g = -GM/r = -(6.67 \times 10^{-11} \times 5.97 \times 10^{24}) / (6.37 \times 10^6) =$ **$-6.26 \times 10^7$ J kg$^{-1}$**

$g = GM/r^2 = (6.67 \times 10^{-11} \times 5.97 \times 10^{24}) / (6.37 \times 10^6)^2 =$ **9.82 N kg$^{-1}$** (close to the standard value of 9.81)

## Orbits and Satellites

For a circular orbit, the gravitational force provides the centripetal force:

$$GMm/r^2 = mv^2/r$$

This gives the orbital speed and Kepler's Third Law:

$$v = \sqrt{(GM/r)}$$

$$T^2 = (4\pi^2/GM)\, r^3$$

**Kepler's Third Law:** $T^2$ is proportional to $r^3$ for any objects orbiting the same central mass.

**Total energy of an orbiting satellite:**

- KE = $\frac{1}{2}mv^2 = GMm/(2r)$
- PE = $-GMm/r$
- Total energy = KE + PE = $-GMm/(2r)$ (always negative for a bound orbit)

As r increases, KE decreases and PE increases (becomes less negative). The total energy also increases (becomes less negative).

## Geostationary Orbits

**Geostationary orbit:** A special orbit where the satellite remains above the same point on the Earth's equator at all times. Requirements:

- Orbital period T = 24 hours (86400 s)
- Orbits directly above the equator
- Orbits in the same direction as Earth's rotation (west to east)

- Orbital radius ≈ 42,200 km from Earth's centre (altitude ≈ 35,800 km)

Uses: satellite TV, communications, weather monitoring.

### Escape Velocity

The minimum velocity needed for an object to escape a gravitational field entirely (reach infinity with zero KE). Found by equating initial KE to the magnitude of the gravitational PE:

$$v_{esc} = \sqrt{(2GM/r)}$$

For Earth's surface: $v_{esc} \approx 11.2$ km $s^{-1}$.

**Exam Tip:** Note that escape velocity is √2 times the orbital velocity at the same radius: $v_{esc} = \sqrt{2} \times v_{orbital}$. A geostationary orbit has T = 24 hours. Using Kepler's Third Law: $r = (GMT^2/(4\pi^2))^{1/3} \approx$ 42,200 km from Earth's centre.

## 2. Electric Fields

**Electric field:** A region of space in which a charged particle experiences a force. Electric field lines go from positive to negative charges. Like charges repel; unlike charges attract.

### Coulomb's Law

$$F = kQ_1Q_2 / r^2 = Q_1Q_2 / (4\pi\varepsilon_0 r^2)$$

- $k = 1/(4\pi\varepsilon_0) = 8.99 \times 10^9$ N $m^2$ $C^{-2}$
- $\varepsilon_0$ = permittivity of free space = $8.85 \times 10^{-12}$ F $m^{-1}$
- F is positive (repulsion) for like charges, negative (attraction) for unlike charges

### Electric Field Strength

**Electric field strength (E):** The force per unit positive charge at a point. E = F/Q. Units: N $C^{-1}$ or V $m^{-1}$.

**Radial field** (point charge):

$$E = kQ / r^2 = Q / (4\pi\varepsilon_0 r^2)$$

**Uniform field** (between parallel plates):

$$E = V / d$$

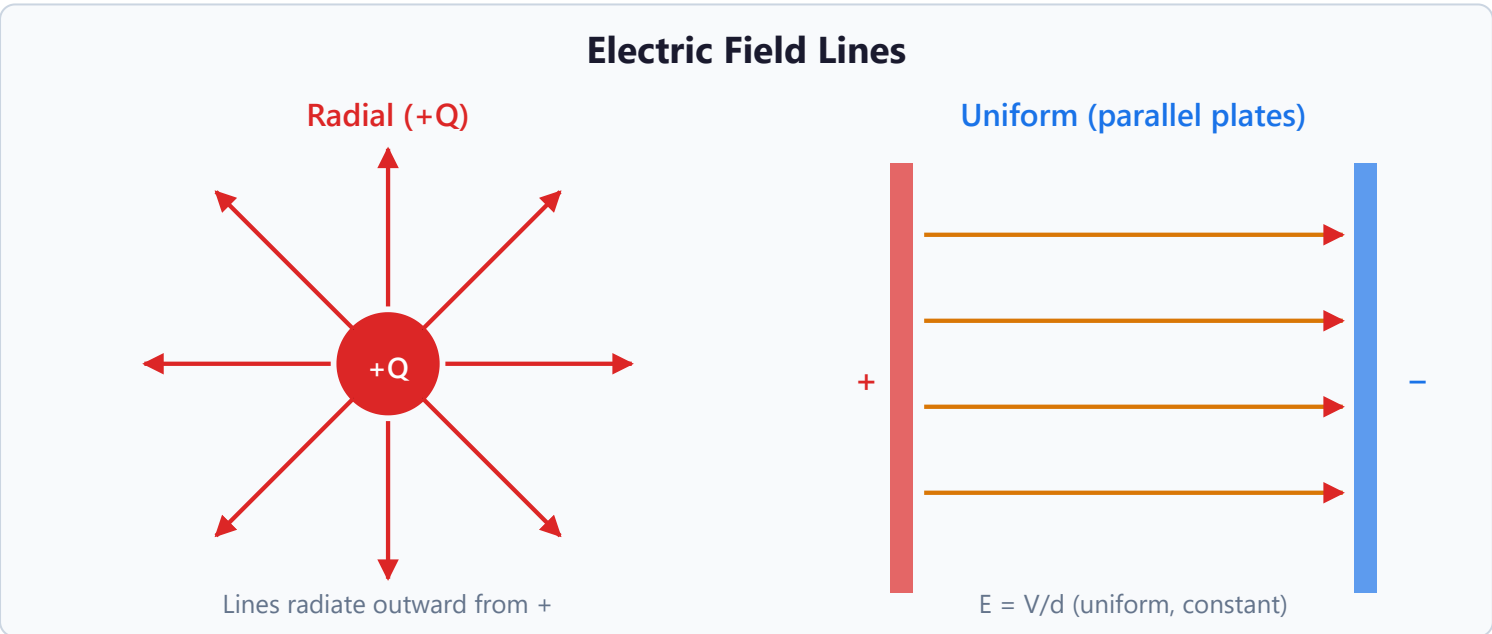
where V = potential difference between the plates and d = plate separation. The field is uniform (constant E) between the plates, and field lines are straight, parallel, and equally spaced from + plate to − plate.

## Electric Potential

**Electric potential ($V_e$):** The work done per unit positive charge in bringing a small positive test charge from infinity to that point. Units: V (or J $C^{-1}$).

$$V_e = kQ / r = Q / (4\pi\varepsilon_0 r)$$

$V_e$ is positive near a positive charge and negative near a negative charge. The relationship $E = -dV_e/dr$ applies (field strength is the negative potential gradient).




**Worked Example:** A proton (charge $+1.60 \times 10^{-19}$ C) is placed between two parallel plates separated by 2.0 cm with a PD of 500 V across them. Calculate (a) the electric field strength, (b) the force on the proton, (c) the acceleration of the proton. ($m_p = 1.67 \times 10^{-27}$ kg)

**Solution:**

(a) $E = V/d = 500 / 0.020 =$ **$25000\ V\ m^{-1}$** $= 2.5 \times 10^4\ N\ C^{-1}$

(b) $F = EQ = 25000 \times 1.60 \times 10^{-19} =$ **$4.0 \times 10^{-15}\ N$**

(c) $a = F/m = 4.0 \times 10^{-15} / 1.67 \times 10^{-27} =$ **$2.4 \times 10^{12}\ m\ s^{-2}$**

This enormous acceleration illustrates why electric fields are so effective at accelerating charged particles.

## Comparison: Gravitational and Electric Fields

| Property | Gravitational | Electric |
|---|---|---|
| Source | Mass | Charge |
| Force law | $F = -GMm/r^2$ | $F = kQ_1Q_2/r^2$ |
| Nature | Always attractive | Attractive or repulsive |
| Field strength | $g = GM/r^2$ ($N\ kg^{-1}$) | $E = kQ/r^2$ ($N\ C^{-1}$) |
| Potential | $V = -GM/r$ (always negative) | $V = kQ/r$ (can be + or −) |
| Field strength ↔ potential | $g = -dV/dr$ | $E = -dV/dr$ |
| Relative strength | Very weak | Very strong (~$10^{36}$ times stronger) |
| Shielding | Cannot be shielded | Can be shielded (Faraday cage) |

**Exam Tip:** Potential is a scalar — add algebraically with sign. Field strength is a vector — add using vector rules. When finding the resultant potential at a point due to multiple charges, just add the values (with signs). For field strength, you must resolve into components and add as vectors. A common exam question asks you to find the point where the resultant field is zero between two charges.

**Worked Example:** Two point charges, $Q_1 = +4.0$ nC and $Q_2 = -2.0$ nC, are separated by 0.30 m. Find the point along the line between them where the electric potential is zero.

**Solution:**

Let the point be at distance x from $Q_1$ and (0.30 − x) from $Q_2$.

$V = kQ_1/x + kQ_2/(0.30 - x) = 0$

$Q_1/x = -Q_2/(0.30 - x)$

$4.0/x = 2.0/(0.30 - x)$

$4.0(0.30 - x) = 2.0x$

$1.2 - 4.0x = 2.0x$

$1.2 = 6.0x$

x = **0.20 m** from $Q_1$ (and 0.10 m from $Q_2$)

## 3. Capacitance

**Capacitance (C):** The charge stored per unit potential difference across a capacitor. C = Q/V. Units: F (farads). 1 F = 1 C $V^{-1}$. Typical values are in the μF ($10^{-6}$), nF ($10^{-9}$), or pF ($10^{-12}$) range.

### Parallel Plate Capacitor

$$C = \varepsilon_0 \varepsilon_r A / d$$

- $\varepsilon_0$ = permittivity of free space = $8.85 \times 10^{-12}$ F $m^{-1}$
- $\varepsilon_r$ = relative permittivity (dielectric constant) of the material between the plates
- A = area of overlap of the plates ($m^2$)
- d = separation of the plates (m)

A dielectric material placed between the plates increases capacitance by a factor of $\varepsilon_r$. The dielectric reduces the electric field inside the capacitor because its molecules become polarised and partially cancel the applied field.

### Capacitors in Series and Parallel

| Arrangement | Rule | Effect |
|---|---|---|

| | | |
|---|---|---|
| Series | $1/C_{total} = 1/C_1 + 1/C_2 + ...$ | Total capacitance decreases (always less than smallest individual C) |
| Parallel | $C_{total} = C_1 + C_2 + ...$ | Total capacitance increases |

**Exam Tip:** Capacitors in series/parallel combine the OPPOSITE way to resistors. Capacitors in series use the reciprocal rule (like resistors in parallel). Capacitors in parallel add directly (like resistors in series). This is because adding capacitors in parallel increases the total plate area, while series decreases effective capacitance.

## Energy Stored in a Capacitor

$$E = ½QV = ½CV^2 = ½Q^2/C$$

The energy stored equals the area under the Q-V graph (which is a straight line through the origin with gradient C). The factor of ½ arises because the average voltage during charging is V/2.

**Worked Example:** A 470 μF capacitor is charged to 12 V. Calculate (a) the charge stored, (b) the energy stored. The capacitor is then disconnected and connected across a 1000 μF uncharged capacitor. Find (c) the final voltage across both capacitors, (d) the energy lost.

**Solution:**

(a) $Q = CV = 470 \times 10^{-6} \times 12 =$ **$5.64 \times 10^{-3}$ C** (5.64 mC)

(b) $E = ½CV^2 = ½ \times 470 \times 10^{-6} \times 12^2 =$ **0.0338 J** (33.8 mJ)

(c) Charge is conserved: $Q_{total} = 5.64 \times 10^{-3}$ C. $C_{total} = 470 + 1000 = 1470$ μF.

$V_{final} = Q/C_{total} = 5.64 \times 10^{-3} / 1470 \times 10^{-6} =$ **3.84 V**

(d) $E_{final} = ½C_{total}V^2 = ½ \times 1470 \times 10^{-6} \times 3.84^2 = 0.0108$ J

Energy lost = 0.0338 − 0.0108 = **0.0230 J** (23.0 mJ, dissipated as heat in the connecting wires)

## Charging and Discharging Through a Resistor

The charge, voltage, and current change exponentially:

**Discharging:**

$$Q = Q_0 e^{-t/RC} \quad V = V_0 e^{-t/RC} \quad I = I_0 e^{-t/RC}$$

**Charging:**

$$Q = Q_0(1 - e^{-t/RC}) \quad V = V_0(1 - e^{-t/RC})$$

Current during charging still decays exponentially: $I = I_0 e^{-t/RC}$ (highest at the start, falls to zero as the capacitor becomes fully charged).

### Time Constant

$$\tau = RC$$

The time constant is the time for the charge (or voltage) to fall to 1/e (≈ 37%) of its initial value during discharge, or to rise to (1 − 1/e) ≈ 63% of its final value during charging. After 5τ, the capacitor is considered fully charged or discharged (< 1% remaining or > 99% charged).

**Exam Tip:** To find τ from a discharge graph, read off the time when the value drops to 37% of the initial value. To linearise the data, plot ln(V) against t — the gradient is −1/RC and the y-intercept is $\ln(V_0)$.

## 4. Magnetic Fields

**Magnetic flux density (B):** A measure of the strength of a magnetic field at a point. It represents the force per unit current per unit length on a conductor carrying current perpendicular to the field. Units: T (tesla). 1 T = 1 kg $s^{-2}$ $A^{-1}$ = 1 Wb $m^{-2}$.

### Force on a Current-Carrying Conductor

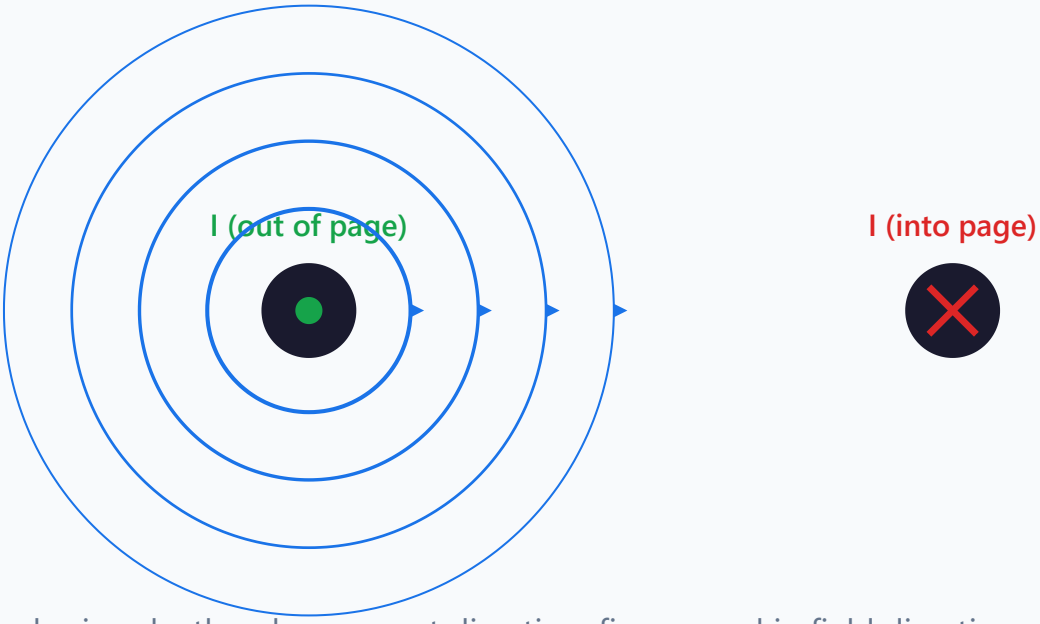
$$F = BIL \sin\theta$$

- F = force on the conductor (N)
- B = magnetic flux density (T)
- I = current through the conductor (A)
- L = length of conductor in the field (m)
- θ = angle between the current direction and the magnetic field direction

- Maximum force when θ = 90° (I perpendicular to B). Zero force when θ = 0° (I parallel to B).

Direction given by **Fleming's left-hand rule**: thuMb = Motion (force), First finger = Field (B), seCond finger = Current (I).

## Magnetic Field Around a Current-Carrying Wire




Right-hand grip rule: thumb = current direction, fingers curl in field direction

### Force on a Moving Charge

$$F = BQv \sin\theta$$

A charged particle moving perpendicular to a uniform magnetic field follows a circular path (the magnetic force provides the centripetal force, always perpendicular to velocity):

$$BQv = mv^2/r \;\Rightarrow\; r = mv / (BQ)$$

The radius of the circular path increases with speed or mass, and decreases with stronger field or larger charge. The magnetic force does no work (F is always perpendicular to v), so the speed and KE remain constant.

**Worked Example:** An electron ($m = 9.11 \times 10^{-31}$ kg, $q = 1.60 \times 10^{-19}$ C) moves at $2.0 \times 10^{7}$ m s$^{-1}$ perpendicular to a magnetic field of 0.10 T. Calculate the radius of its circular path.

**Solution:**

$r = mv/(BQ) = (9.11 \times 10^{-31} \times 2.0 \times 10^{7}) / (0.10 \times 1.60 \times 10^{-19})$

$r = (1.822 \times 10^{-23}) / (1.60 \times 10^{-20}) = \mathbf{1.14 \times 10^{-3}\ m}$ (1.1 mm)

## Magnetic Flux and Flux Linkage

**Magnetic flux (Φ):** The total magnetic field passing through an area. $\Phi = BA\cos\theta$, where θ is the angle between B and the normal to the surface. Units: Wb (webers). 1 Wb = 1 T $m^2$.

**Flux linkage (NΦ):** For a coil of N turns, flux linkage = $N\Phi = BAN\cos\theta$. Units: Wb (or Wb-turns).

## Electromagnetic Induction

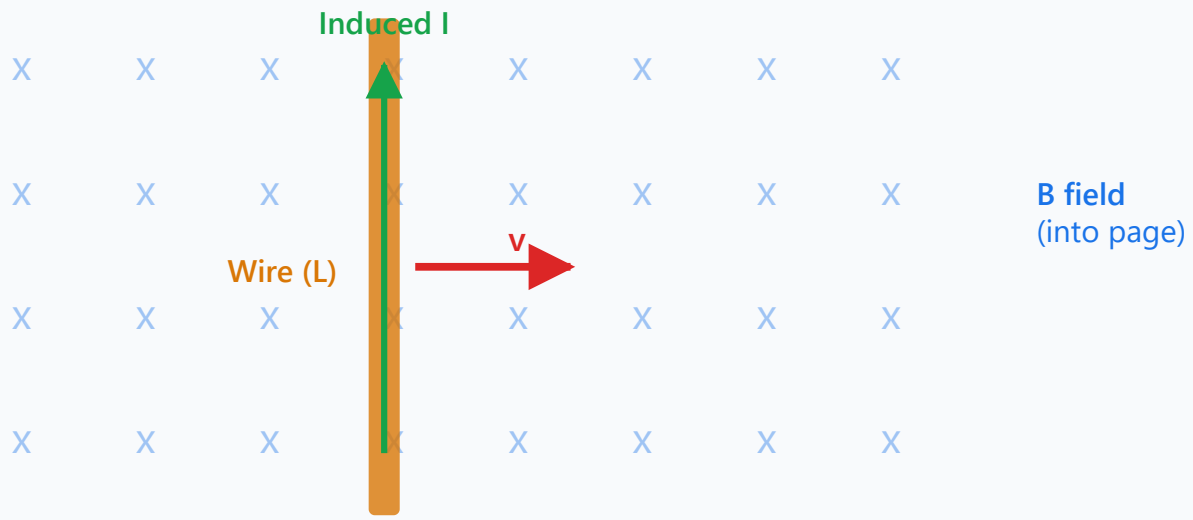
**Faraday's Law:** The magnitude of the induced EMF equals the rate of change of flux linkage:
$\varepsilon = -N \times d\Phi/dt = -d(N\Phi)/dt$

**Lenz's Law:** The direction of the induced EMF (and current) is such that it opposes the change in flux that produced it. This is the reason for the negative sign in Faraday's Law and is a consequence of conservation of energy.

For a straight wire of length L moving at velocity v perpendicular to a uniform field B:

$$\varepsilon = BLv$$

## Electromagnetic Induction




EMF = BLv (wire length L, speed v, field B)

## EMF in a Rotating Coil (AC Generator)

A coil of N turns and area A rotating at angular velocity ω in a uniform field B generates a sinusoidal EMF:

$$\varepsilon = BAN\omega \sin(\omega t)$$

Peak EMF: $\varepsilon_0 = BAN\omega$ (when the coil plane is parallel to the field, i.e. the flux linkage is changing most rapidly).

## Alternating Current

For a sinusoidal AC supply:

$$V_{rms} = V_0 / \sqrt{2} \quad I_{rms} = I_0 / \sqrt{2}$$

$V_{rms}$ and $I_{rms}$ are the root-mean-square values — the DC equivalent that would deliver the same power. UK mains: $V_{rms}$ = 230 V, f = 50 Hz (so $V_0 \approx 325$ V, T = 20 ms).

Mean power: $P = I_{rms}V_{rms} = I_{rms}^2R = V_{rms}^2/R = ½I_0V_0$

## Transformers

$$N_s/N_p = V_s/V_p$$

For an ideal (100% efficient) transformer: $V_pI_p = V_sI_s$ (input power = output power).

- **Step-up transformer:** $N_s > N_p$, so $V_s > V_p$ and $I_s < I_p$
- **Step-down transformer:** $N_s < N_p$, so $V_s < V_p$ and $I_s > I_p$

**Transformer operation:** AC in the primary coil produces a changing magnetic flux in the soft iron core. This changing flux links to the secondary coil and, by Faraday's law, induces an EMF. Transformers only work with AC (or changing current), not DC.

**Energy losses in real transformers:**

| Loss Mechanism | Cause | Reduction Method |
|---|---|---|
| Copper losses ($I^2R$ heating) | Resistance of the coil windings | Use thick, low-resistance copper wire |
| Eddy current losses | Induced currents in the iron core dissipate energy as heat | Laminate the core (thin insulated layers) |
| Hysteresis losses | Energy lost in repeatedly magnetising and demagnetising the core | Use a soft magnetic material (low hysteresis) |
| Flux leakage | Not all flux from primary links to secondary | Wind coils on top of each other; use a closed core |

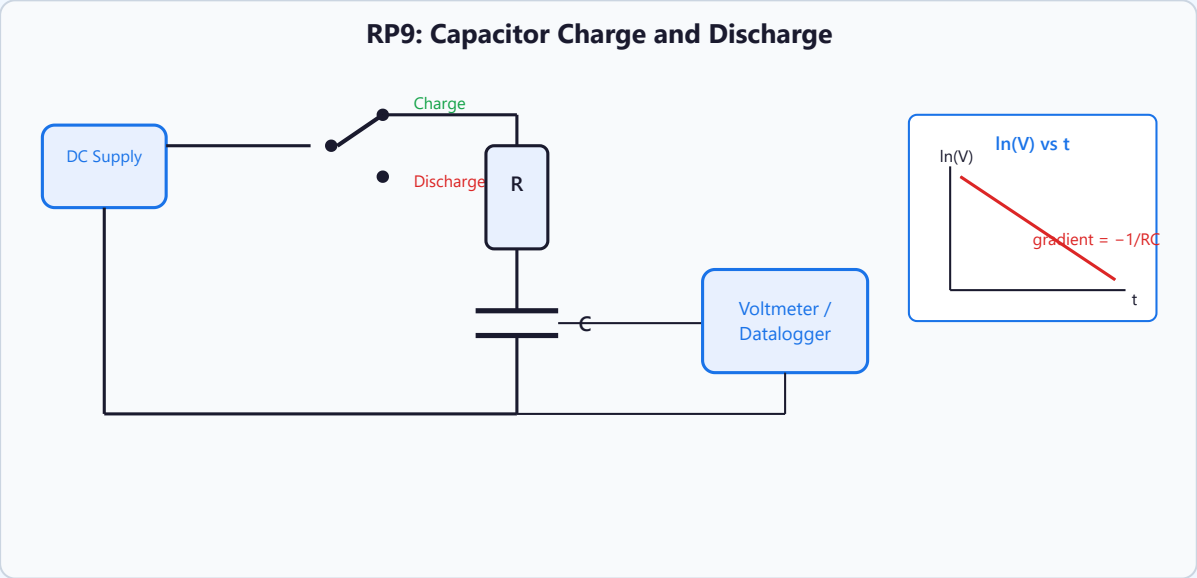
**Exam Tip:** High-voltage transmission reduces power loss in cables. $P_{loss} = I^2R_{cable}$. By stepping up the voltage (and thus reducing I for the same power P = IV), the $I^2R$ losses are dramatically reduced. Doubling V halves I and quarters the power loss. The UK National Grid operates at up to 400 kV for this reason.

## Summary of Key Equations

| Topic | Key Equations |
|---|---|
| Gravitation | $F = GMm/r^2$; $g = GM/r^2$; $V = -GM/r$; $v_{esc} = \sqrt{(2GM/r)}$ |
| Orbits | $v = \sqrt{(GM/r)}$; $T^2 = (4\pi^2/GM)r^3$ |

| | |
|---|---|
| Electric fields | $F = kQ_1Q_2/r^2$; $E = kQ/r^2$; $V = kQ/r$; $E = V/d$ (uniform) |
| Capacitance | $C = Q/V$; $E = ½CV^2$; $\tau = RC$; $Q = Q_0e^{-t/RC}$ |
| Magnetic force | $F = BIL \sin\theta$; $F = BQv \sin\theta$; $r = mv/(BQ)$ |
| Induction | $\varepsilon = -N\, d\Phi/dt$; $\Phi = BA \cos\theta$; $\varepsilon = BLv$ |
| Transformers | $N_s/N_p = V_s/V_p$; $V_{rms} = V_0/\sqrt{2}$ |

## Required Practical 9: Capacitor Charge and Discharge




### Apparatus

- Capacitor (e.g. 470 µF or 1000 µF electrolytic)
- Resistor (e.g. 10 kΩ or 100 kΩ — chosen so $\tau = RC$ is several seconds)
- DC power supply (e.g. 6 V)
- Two-way switch (DPDT) for charge/discharge
- Voltmeter or datalogger with voltage sensor (across the capacitor)
- Stopwatch (if not using a datalogger)

### Method

1. Connect the circuit as shown. Set the switch to the charge position and let the capacitor fully charge to $V_0$.
2. **Discharging:** Switch to discharge. Record voltage V at regular time intervals (e.g. every 5 s, or use a datalogger for continuous recording).
3. Continue recording until V drops below 10% of $V_0$.
4. Plot V vs t (gives an exponential decay curve). Plot ln(V) vs t (gives a straight line with gradient = −1/RC).
5. Calculate τ = RC = −1/gradient and compare with the theoretical value.

## Safety

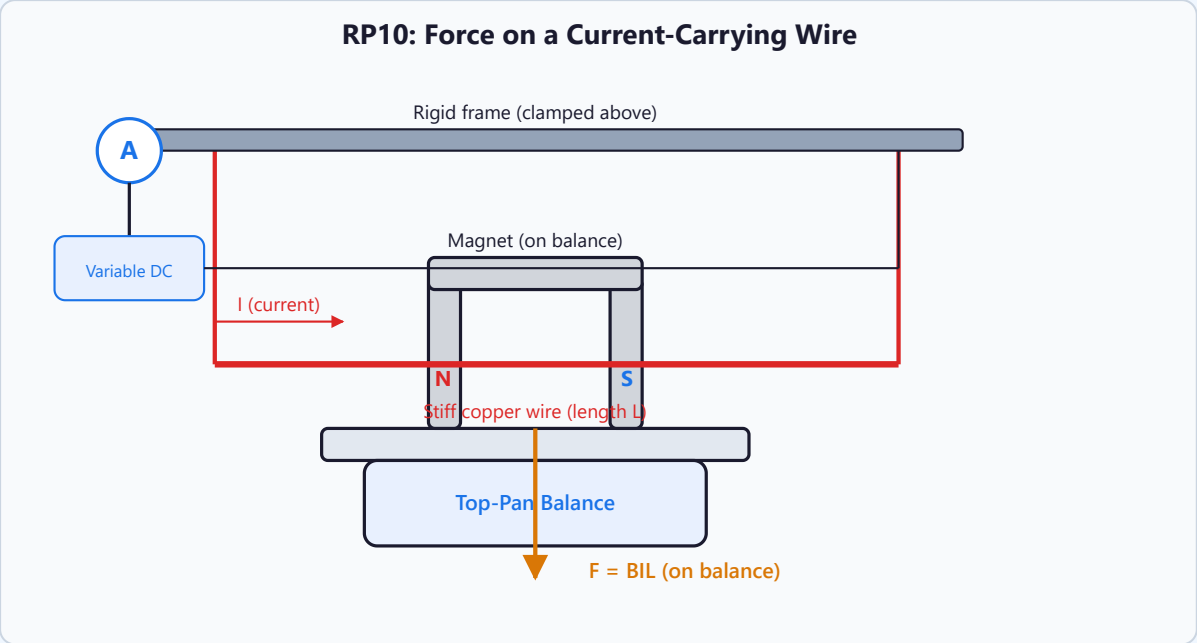
- Use low voltages (≤ 6 V). Ensure electrolytic capacitors are connected with correct polarity.
- Discharge the capacitor fully before handling. Large capacitors can deliver a painful shock.

## Sources of Error & Improvements

| Source of Error | Type | Improvement |
|---|---|---|
| Voltmeter draws current, reducing effective R | Systematic | Use a high-impedance voltmeter or datalogger |
| Timing is difficult if RC is too small | Random | Choose R and C to give τ = 10–60 s; use a datalogger |
| Component tolerances differ from labelled values | Systematic | Measure R and C independently; compare theoretical and experimental τ |
| Leakage current in the capacitor | Systematic | Use a high-quality capacitor; take readings quickly |

**Exam Tip:** The ln(V) vs t graph linearises the exponential. If it is a straight line, the discharge is exponential, confirming $V = V_0e^{-t/RC}$. The time constant can also be read directly from the V vs t graph as the time at which $V = 0.37V_0$.

# ⚗ Required Practical 10: Force on a Current-Carrying Wire




## Apparatus

Strong horseshoe magnet, stiff copper wire (known length L), top-pan balance (0.01 g), variable DC power supply, ammeter, retort stand and clamps, connecting leads.

## Method

1. Place the magnet on the top-pan balance. Zero (tare) the balance.
2. Clamp a rigid wire frame so the horizontal section sits in the gap between the poles, perpendicular to B.
3. Set a current I and read the apparent mass change Δm. Calculate force: $F = \Delta m \times g$.
4. Vary current in steps. Record Δm and F at each current.
5. Plot F vs I. Gradient = BL, so B = gradient/L.

## Safety

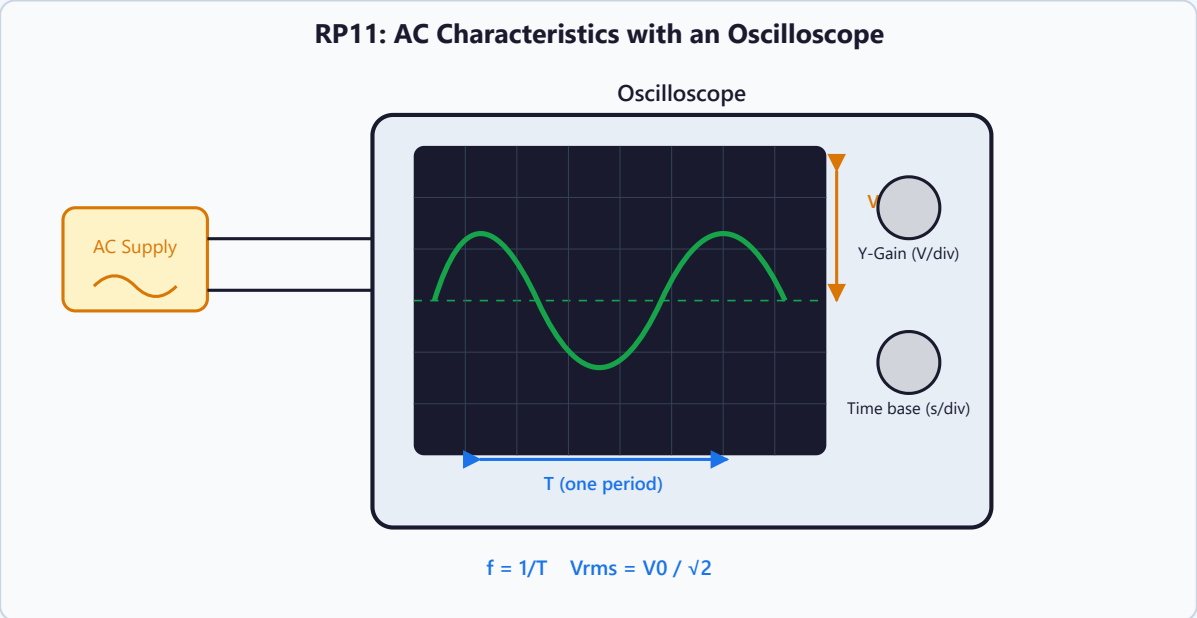
- Keep currents moderate to avoid overheating. Switch off between readings.
- Strong magnets can pinch fingers. Keep magnets away from electronic devices.

## Sources of Error & Improvements

| Source of Error | Type | Improvement |
|---|---|---|
| Wire not perpendicular to B | Systematic | Align carefully with a set square |
| Non-uniform field | Systematic | Keep wire centred in the gap; use a strong magnet |
| Balance fluctuations | Random | Shield from draughts; average readings |
| Connecting leads exert forces | Systematic | Arrange leads symmetrically away from magnet |

**Exam Tip:** The balance method uses Newton's third law: the force on the wire (F = BIL) produces an equal and opposite reaction force on the magnet, detected by the balance. If the wire is pushed up, the magnet is pushed down, increasing the balance reading.

## Required Practical 11: AC Supply with an Oscilloscope




### Apparatus

Oscilloscope (CRO or digital), AC signal generator or low-voltage AC supply, connecting leads.

## Method

1. Connect the AC supply to the Y-input of the oscilloscope.
2. Adjust the time base (s/div) until 1–3 complete cycles are visible.
3. Adjust the Y-gain (V/div) so the wave fills most of the screen.
4. Measure $V_0$: count divisions from the centre line to the peak × Y-gain setting.
5. Measure T: count divisions for one complete cycle × time base setting.
6. Calculate f = 1/T and $V_{rms} = V_0/\sqrt{2}$.

## Safety

- Use a low-voltage AC supply (signal generator, typically < 10 V).
- Do not connect the oscilloscope directly to mains voltage.

## Sources of Error & Improvements

| Source of Error | Type | Improvement |
|---|---|---|
| Difficult to read exact divisions (trace width) | Random | Use cursor measurements on a digital oscilloscope; maximise Y-gain |
| Calibration of Y-gain or time base may be off | Systematic | Calibrate using a known signal; check with multimeter |
| AC frequency may drift | Random | Use a stable signal generator; take multiple readings |

**Exam Tip:** $V_{rms}$ is the DC equivalent voltage that delivers the same power: $V_{rms} = V_0/\sqrt{2}$. The oscilloscope displays $V_0$ (peak), while an AC multimeter reads $V_{rms}$. For UK mains: $V_{rms}$ = 230 V, $V_0$ = 325 V, f = 50 Hz, T = 20 ms.